

Name: _____

Date: _____

Use the following equation for uncertainty calculations.

$$E = \frac{3mgLV_iG(GF)}{wt^2\delta V_o}.$$

See the experimental handout for variable description. Show all your derivations and hand calculations here for one selected weight.

Table 1: Table to calculate uncertainty in Youngs modulus (E) using error propagation approach. Fill in the separate tables for m = 0.5, 1.0, and 1.5 kg.

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[\delta V_o]_{avg}$ (v)
Variable mean value	0.5								
Vairable uncertainty									
Uncertainty propagated to E (GPa)									
Total uncertainty in E (GPa)									

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[\delta V_o]_{avg}$ (v)
Variable mean value	1.0								
Vairable uncertainty									
Uncertainty propagated to E (GPa)									
Total uncertainty in E (GPa)									

	m (kg)	g (m/s^2)	L(m)	t(m)	w(m)	GF	G	V_i (v)	$[\delta V_o]_{avg}$ (v)
Variable mean value	1.5								
Vairable uncertainty									
Uncertainty propagated to E (GPa)									
Total uncertainty in E (GPa)									